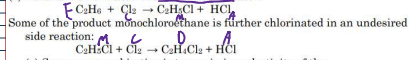


Question 1

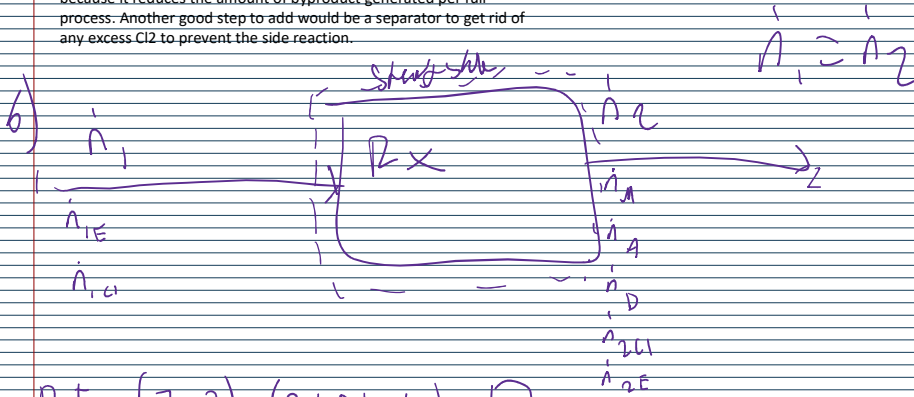
Monday, October 20, 2025 9:05 AM

(1) Ethane is chlorinated in a continuous reactor:



- (a) Suppose your objective is to maximize selectivity of the monochloroethane production relative to the dichloroethane production. Would you design the reactor for a high or low conversion of ethane? Explain your answer. What additional processing steps would almost certainly be necessary to make the process economical?
- (b) Draw and label a flowchart, assuming that the reactor feed contains only the two reactants. Use a degree-of-freedom analysis based on atomic species balances to determine how many process variable values must be specified for you to be able to calculate the remainder.
- (c) The reactor is designed to yield a 15% conversion of ethane and a selectivity of 14 moles of $\text{C}_2\text{H}_5\text{Cl}$ for every mole of $\text{C}_2\text{H}_4\text{Cl}_2$. Assume all of the chlorine gas is reacted. Calculate the feed ratio (moles Cl_2 /moles C_2H_6) and the fractional yield of monochloroethane.
- (d) Suppose the reactor is built and started up and the conversion is only 14%. Chromatographic analysis shows that there is no Cl_2 in the product, but another species of molecular weight higher than that of dichloroethane is present. Offer a likely explanation of the results.

A) Designing the reactor for a low conversion rate would be better, because it reduces the amount of byproduct generated per full process. Another good step to add would be a separator to get rid of any excess Cl_2 to prevent the side reaction.



$$\text{DoF} = (7+2) - (0+0+0+5) = 4 \text{ unknown}$$

need:

- conversion rate
- selectivity
- basis or any other assumptions

What is the decision made to increase selectivity?

c) 15% conversion $E = 100\% \cdot \frac{n_{1E} - n_{2E}}{n_{1E}}$

$$n_{2E} = 0.85 n_{1E}$$

$$n_{2Cl_2} = 0$$

$$f_{\text{selective}} = \frac{14 \text{ mol } M}{1 \text{ mol } D}$$

Assume basis: $n_{1E} = 100 \frac{\text{mol}}{\text{s}}$

$$\text{DoF} = (7+2) - (1+0+3+5) = 0$$

$$n_{2E} = 85 \frac{\text{mol } E}{\text{s}}$$

Atomic Balancing

C: $2E_1 = 2M + 2D + 2(0.85E)$

H: $6E_1 = 5M + 4D + 2A + 6(0.85E)$

$2D \geq 2M + 2D \rightarrow 30 = 30 \text{ mol}$

$$2D \cdot \frac{14 \text{ mol}}{1 D} = 28 \text{ mol}$$

$$4D \cdot \frac{14 \text{ mol}}{1 D} = 56 \text{ mol}$$

$$M = 1 \frac{\text{mol } M}{\text{s}} = \dot{n}_M$$

H: 0.1 ...

$$30 = 2M + 2D \rightarrow 30 = 20M$$

$$90 = 5M + 4D + 2A \rightarrow 90 = 60M + 2A$$

$$M = \frac{1 \text{ mol M}}{s} = \dot{n}_M$$

$$A = \frac{15 \text{ mol A}}{s} = \dot{n}_A$$

$$D = \frac{1 \text{ mol M}}{s} \cdot \frac{1 \text{ mol D}}{14 \text{ mol M}} = \frac{0.071 \text{ mol D}}{s} = \dot{n}_D$$

$$\dot{n}_1 = 100 \frac{\text{mol E}}{s} + \dot{n}_{\text{ICI}}$$

$$\dot{n}_2 = 85 \frac{\text{mol E}}{s} + 1 \text{ mol M} + 15 \frac{\text{mol A}}{s} + 0.071 \frac{\text{mol D}}{s} = 101.071 \frac{\text{mol}}{s}$$

$$100 + \dot{n}_{\text{ICI}} = 101.071$$

$$\dot{n}_{\text{ICI}} = 1.071 \frac{\text{mol Cl}}{s}$$

$$\frac{10}{25}$$

$$\frac{1.071 \frac{\text{mol Cl}}{s}}{100 \frac{\text{mol E}}{s}} = 0.01071 \frac{\text{mol Cl}}{\text{mol E}}$$

$$f_{\text{field}} = \frac{1 \text{ mol M}}{100 \text{ mol E}} = 0.01 \frac{\text{mol M}}{\text{mol E}} \quad \text{X}$$

D) One possible explanation for this could be that the dichloroethane has been chlorinated once more in a second unwanted reaction, resulting in trichloroethane.

Question 2

Monday, October 20, 2025 9:05 AM

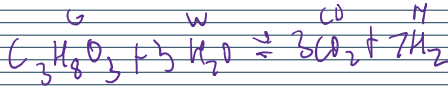
(2) Murphy 4.40 (You are given $K_a = 55 \text{ atm}^6$ which is the equilibrium constant. Under these conditions K_a is written in terms of partial pressures

so that $K_a = \frac{(P_C)^2 (P_H)^7}{P_G^3 (P_W)^3}$
 where P_G is the partial pressure of the CO , P_H for the H_2 , P_G for the glycerol ($\text{C}_3\text{H}_8\text{O}_3$), and P_W for the water (H_2O). This should explain why K_a has units of atm^6 . The partial pressures are related to the mole fraction by $P_i = x_i P$ where P_i is the partial pressure for species i , x_i is the mole fraction for species i , and P is the total pressure.)

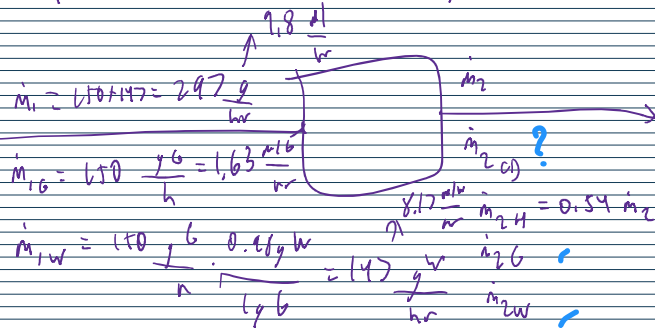
gas

$$P = 1.2 \text{ atm}$$

$$T = 200^\circ\text{C}$$



$$\frac{5 \text{ mol W}}{1 \text{ mol G}} \cdot \frac{18 \text{ g W}}{1 \text{ mol W}} \cdot \frac{1 \text{ mol G}}{92 \text{ g G}} = \frac{0.98 \text{ g W}}{1 \text{ g G}}$$



$$\text{DoF} = (6+1) - (1+1+2+4) = -1 \text{ over specified}$$

Species Balances

$$x_{2G} \dot{n}_2 = 1.63 \text{ g/hr} \rightarrow (x_{2G} + 0.077) \dot{n}_2 = 1.63$$

$$x_{2W} \dot{n}_2 = 8.17 \text{ g/hr} \rightarrow (x_{2W} + 0.23) \dot{n}_2 = 8.17$$

$$x_{2CO} \dot{n}_2 = 3 \text{ g/hr} \rightarrow x_{2CO} \dot{n}_2 = 0.23 \dot{n}_2$$

$$0.54 \dot{n}_2 = 7 \text{ g/hr} \rightarrow \dot{n}_2 = 0.077 \dot{n}_2 \quad [x_{2G} = 0.63]$$

$$x_{2G} + x_{2W} = 0.23$$

$$x_{2G} = 0.23 - x_{2W}$$

$$\frac{0.23 - x_{2W} + 0.077}{x_{2W} + 0.23} = 0.20375$$

$$\frac{0.307 - x_{2W}}{x_{2W} + 0.23} = 0.20375$$

mass balance:

$$\dot{m}_1 = \dot{m}_2 = 297 \text{ g/hr}$$

$$0.307 - x_{2W} = 0.20375 x_{2W} + 0.097$$

$$0.26 = 1.70375 x_{zw}$$

$$x_{zw} = 0.26$$

$$x_{zg} = 0.014$$

$$P_2 = (0.23(44) + 0.54(22) + 0.26(48) + 0.014(72)) \frac{\text{atm}}{\text{hr}} = 18.14 \frac{\text{atm}}{\text{hr}}$$

$$f_{\text{conversion}} = \frac{(1.63 - 0.014(18.14)) \cdot 6}{1.63 \cdot 6} = 0.84 = 84\%$$

$$p_i = x_i P$$

$$p_{wv} = 0.23(1.2 \text{ atm}) = 0.276 \text{ atm}$$

$$p_{wv} = 0.26(1.1) = 0.2542 \text{ atm}$$

$$p_g = 0.014(1.1) = 0.0168 \text{ atm}$$

$$p_H = 0.54(1.1) = 0.648$$

$$K_a = \frac{(0.276 \text{ atm})^3 (0.648)^7}{(0.0168 \text{ atm})(0.2542 \text{ atm})^3} = 3.73 \text{ atm}^6$$

not in equilibrium

1) One way to do better is to increase the conversion rate of the glycine. We can probably do this by increasing the pressure on the side of the reactants to force a greater output of products via Le Chatelier's Principle.

2) Another possible solution could be to simply increase the temperature of the system to catalyze the reaction and make it occur more often.

20/25

need to
re-calculate

x

$$f_{\text{conversion}} = 1 \approx 100\%$$

Assume that equilibrium means an almost 100% conversion of glycerine

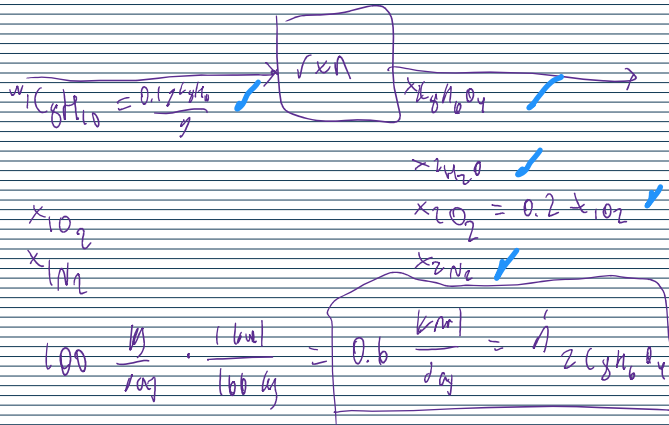
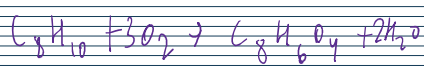
-> x_{H_2} is based purely on stoichiometry

$$\frac{1.63 \text{ mol G}}{1 \text{ h}} = \dot{n}_{1G}$$

$$\frac{7 \text{ mol H}_2}{1 \text{ mol G}} \cdot \frac{1 \text{ h}}{18.14 \text{ mol}} = \dot{n}_2$$

$$\frac{0.63 \text{ mol H}_2}{0.57 \text{ mol}} = x_{H_2}$$

OK



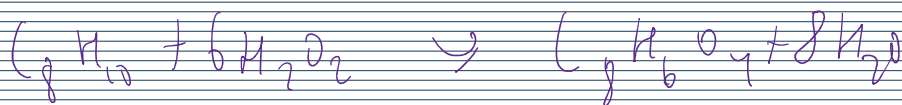
$$0.6 \frac{\text{kmol}}{\text{day}} \cdot \frac{1 \text{ kmol } C_8H_{10}}{1 \text{ kmol } TA} = 0.6 \frac{\text{kmol } C_8H_{10}}{\text{day}} = \dot{n}_1(C_8H_{10}) \rightarrow 0.6 \cdot (100) = \underline{63.6 \frac{\text{kg } C_8H_{10}}{\text{day}}}$$

$$0.6 \frac{\text{kmol } TA}{\text{day}} \cdot \frac{3 \text{ kmol } O_2}{1 \text{ kmol } TA} = 1.8 \frac{\text{kmol } O_2}{\text{day}} \cdot 1.2 = 2.16 \frac{\text{kmol } O_2}{\text{day}} = \dot{n}_{O_2} \rightarrow 2.16(32) = \underline{69.1 \frac{\text{kg } O_2}{\text{day}}}$$

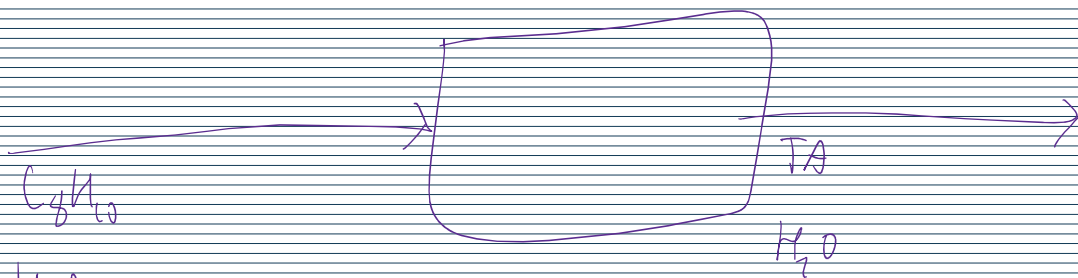
Atom economy:

$$= \frac{106 \text{ kg } TA}{(106 \text{ kg } C_8H_{10}) + 3 \text{ mol } O_2 \left(\frac{32 \text{ kg } O_2}{\text{mol}} \right)} = 0.82 = \boxed{82\%} \quad \checkmark$$

Connet's Process



$$\frac{10 \text{ g } TA}{\text{hr}} \cdot \frac{1 \text{ mol } TA}{106 \text{ g } TA} = 0.06 \frac{\text{mol } TA}{\text{hr}} = \dot{n}_{TA}$$



$$0.7 = \frac{\dot{n}_{TA \text{ out}}}{\dot{n}_{C_8H_{10} \text{ in}}} = \frac{0.06}{\dot{n}_{C_8H_{10} \text{ in}}}$$

$$\dot{n}_{C_8H_{10}} = \frac{0.086 \text{ mol } C_8H_{10}}{\text{hr}}$$

$$\frac{0.086 \text{ mol } C_8H_{10}}{\text{hr}} \cdot \frac{6 \text{ mol } C_8H_{10}}{1 \text{ mol } C_8H_{10}} = 0.52 \frac{\text{mol } C_8H_{10}}{\text{hr}}$$

100 g TA

$$\text{Atom economy} = \frac{(606 \text{ g } C_8H_{10}) / 6 \text{ mol } H_2O_2 \left(34 \frac{\text{g}}{\text{mol}} H_2O_2 \right)}{(606 \text{ g } C_8H_{10}) / 6 \text{ mol } H_2O_2 \left(34 \frac{\text{g}}{\text{mol}} H_2O_2 \right)}$$

$$= 0.54 = \boxed{54\%}$$

25/25

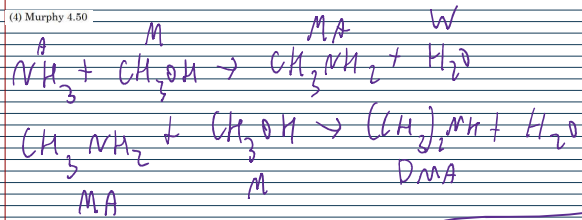
Based on the fact that Connie's process has a decreased atom economy and total yield of TA, I would not pursue Connie's idea further, as it will end up being more costly.



Question 4

Monday, October 20, 2025 9:05 AM

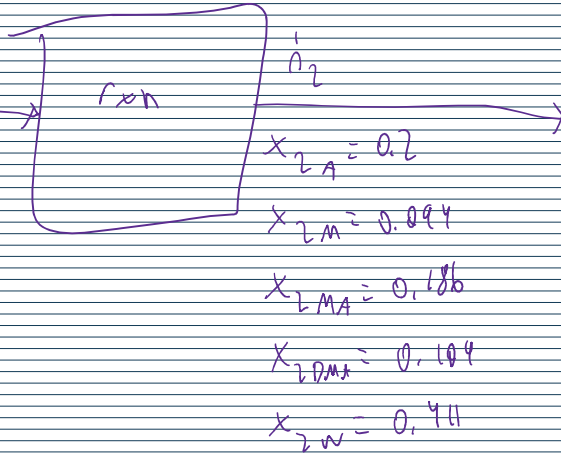
(4) Murphy 4.50



$$\dot{n}_1 = 200 \frac{gm}{hr}$$

$$\dot{n}_{1M} = 100 \frac{gm}{hr}$$

$$\dot{n}_{1A} = 100 \frac{gm}{hr}$$



Atomic Balance

$$C: \dot{n}_{1M} = x_{2MA} \dot{n}_2 + 2x_{2DMA} \dot{n}_2 + x_{2M} \dot{n}_2$$

$$H: 3\dot{n}_{1A} + 9\dot{n}_{1M} = 5x_{2MA} \dot{n}_2 + 2x_{2W} \dot{n}_2 + 7x_{2DMA} \dot{n}_2 + 2x_{2M} \dot{n}_2$$

$$O: \dot{n}_{1M} = x_{2W} \dot{n}_2$$

$$N: \dot{n}_{1A} = x_{2MA} \dot{n}_2 + x_{2DMA} \dot{n}_2 + x_{2A} \dot{n}_2$$

$$\dot{n}_{1M} = \dot{n}_{1A}$$

$$x_{2MA} \dot{n}_2 + 2x_{2DMA} \dot{n}_2 + x_{2M} \dot{n}_2 = x_{2MA} \dot{n}_2 + x_{2DMA} \dot{n}_2 + x_{2A} \dot{n}_2$$

$$x_{2DMA} = x_{2A} - x_{2M} \quad \text{theoretical} \quad \text{experimental}$$

$$x_{2DMA} = (0.2 - 0.094) = 0.106 \approx 0.104$$

The lab analysis is reasonable, as the theoretical and experimental x_{2DMA} values are close.

Species Balance

$$M: 0.094 \dot{n}_2 = 100 - \dot{G}_1 - \dot{G}_2 \quad \dot{n}_1 = 200 \frac{gm}{hr}$$

$$\dot{m}_1 = \dot{m}_2$$

$$M: 0.094 \dot{n}_2 = 100 - \dot{G}_1 - \dot{G}_2 \quad \dot{m}_1 = \dot{m}_2$$

$$A: 0.2 \dot{n}_2 = 100 - \dot{G}_1$$

$$MA: 0.186 \dot{n}_2 = \dot{G}_1 - \dot{G}_2 \quad \dot{m}_1 = \left(\frac{100 \frac{\text{g mol}}{\text{hr}} \cdot 32 \frac{\text{g mol}}{\text{g mol}}}{1 \frac{\text{g mol}}{\text{mol}}} \right) + \left(\frac{100 \frac{\text{g mol}}{\text{hr}} \cdot 17 \frac{\text{g mol}}{\text{g mol}}}{1 \frac{\text{g mol}}{\text{mol}}} \right)$$

$$DMA: 0.109 \dot{n}_2 = \dot{G}_2$$

$$W: 0.411 \dot{n}_2 = \dot{G}_1 + \dot{G}_2 \quad \dot{m}_1 = 7900 \frac{\text{g}}{\text{hr}} = \dot{m}_2$$

$$\dot{n}_2 = 7900 \frac{\text{g}}{\text{hr}} \cdot \frac{1 \frac{\text{g mol}}{\text{mol}}}{(0.094)(32) + (0.2)(17) + (0.186)(31) + (0.109)(21) + (0.411)(18)} \text{ g}$$

$$\dot{n}_2 = 200.19 \frac{\text{g mol}}{\text{hr}}$$

$$a) \left\{ \text{conversion, M} = \frac{100 \frac{\text{g mol}}{\text{hr}} - 200.19 \frac{\text{g mol}}{\text{hr}} (0.094 \frac{\text{g mol}}{\text{g mol}})}{100 \frac{\text{g mol}}{\text{hr}}} = 0.812 = 81.2\% \right. \quad \checkmark$$

$$b) \left\{ \text{yield, MA from M} = \frac{200.19 \frac{\text{g mol}}{\text{hr}} (0.186 \frac{\text{g mol}}{\text{g mol}})}{100 \frac{\text{g mol}}{\text{hr}}} = 0.372 = 37.2\% \right. \quad \checkmark$$

$$c) \left\{ \text{selectivity, MA on M} = \frac{200.19 \frac{\text{g mol}}{\text{hr}} (0.186 \frac{\text{g mol}}{\text{g mol}})}{100 \frac{\text{g mol}}{\text{hr}} - 200.19 \frac{\text{g mol}}{\text{hr}} (0.094 \frac{\text{g mol}}{\text{g mol}})} = 0.459 = 45.9\% \right. \quad \checkmark$$

$$\dot{G}_1 = \dot{r}_1 \quad \dot{G}_2 = \dot{r}_2$$

$$e) 0.109 \left(200.19 \frac{\text{g mol}}{\text{hr}} \right) = \dot{G}_2 = \dot{r}_2$$

$$\dot{r}_2 = 21.8 \frac{\text{g mol}}{\text{hr}}$$

$$d) 0.2 \left(200.19 \frac{\text{g mol}}{\text{hr}} \right) = 100 - \dot{G}_1$$

$$\dot{G}_1 = \dot{r}_1 = 60 \frac{\text{g mol}}{\text{hr}}$$

$$K_1 = 7.0 \quad K_1 = \frac{X_{MA} X_W}{X_A X_M} = \frac{(0.186)(0.21)}{(0.2)(0.024)} = 7.07$$

$$K_2 = 2.5$$

$$K_2 = \frac{X_{DMA} X_W}{X_{MA} X_M} = \frac{(0.109)(0.21)}{(0.186)(0.024)} = 2.56$$

The calculated values are close enough, so the effluent gas is likely at equilibrium

$$\frac{K_1}{K_2} = \frac{(X_{MA})^2}{X_A X_{DMA}}$$

$$X_{MA} = \sqrt{\frac{K_1}{K_2} \cdot X_A \cdot X_{DMA}}$$

Ratio of K1/K2:
Used as function in Excel sheet for Solver

Chose to solve (b): ratio of MA/DMA (book allows for choice)

X_A in	X_M in	X_A out	X_M out	X_MA out	X_DMA out	X_W out	K_1	Target K_1	K_2	Target K_2	M NH3	M CH3OH	M CH3NH2	M (CH3)2NH	M H2O
0.983145557	0.01685444	0.184831	0.003169	0.002957	2.96E-05	0.792159	4.000000	4	2.500000	2.5	17	32	31	45	18 g/gmol
n_A in	n_M in	n_A out	n_M out	n_MA out	n_DMA out	n_W out		k1/k2	k1/k2 * X_A out						
196.6291113	3.37088868	51.47221	0.822408	0.823556	0.008236	220.6019		1.6	0.295730183						
n1 (gmol/hr)	n2 (gmol/hr)			Assumptions: Percent conversion of M (81.2%) is consistent Equilibrium of reactor											
200	278.481987														
m1 = m2 (g/hr)															
4900															
A:M feed ratio	MA/DMA														
58.33153507	99.99997														

In general, increasing the ratio of MA/DMA requires maximizing the ratio of the A/M feeds. There needs to be as much NH3 as possible in the feed, with as little CH4 as possible.

Ratio MA:DMA out ≈ 100 mol MA/mol DMA
- This was the lowest concentration of DMA out possible without resulting in a division by 0

Ratio A:M in ≈ 58.33 mol A/mol M